

COURSE TITLE : WIRELESS AND ADHOC NETWORKS
COURSE CODE : 5263
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Wireless Networks	13
2	Adhoc Networks	13
3	Adhoc Routing	13
4	Transport layer	13

COURSE GENERAL OUTCOMES:

Sl.	G.O	On completion of this course the student will be able to :
1	1	Understand the concepts of Wireless Networks
2	1	Understand wireless ad-hoc networks concepts
3	1	Comprehend the MAC protocols in Ad-hoc networks
	2	Understand the routing process and protocols in Ad-hoc networks
4	1	Understand the transport layer issues in ad-hoc networks
	2	Understand wireless sensor networks

SPECIFIC OUTCOMES:

MODULE – I: Wireless Networks

1.1 Understand the concepts of Wireless Networks

- 1.1.1 Define Wireless LAN –IEEE 802.11 standards.
- 1.1.2 Discuss the Physical layer structure of WLANs.
- 1.1.3 Summarise the MAC sub-layer and MAC Management sub-layer.
- 1.1.4 Explain the concepts of Wireless Security.

- 1.1.5 Discuss WEP and WPA.
- 1.1.6 Differentiate wired LAN and Wireless LAN
- 1.1.7 List issues in wireless LAN
- 1.1.8 Summarize security issues in wireless networks
- 1.1.9 Specify different components of IEEE 802.11 network
- 1.1.10 Explain IEEE 802.11 services
- 1.1.11 Compare IEEE Access point services and station services
- 1.1.12 Differentiate infrastructure based LAN and Adhoc LAN
- 1.1.13 Compare CSMA/CD and CSMA/CA
- 1.1.14 Differentiate frequency hopping spread spectrum and direct sequence spread spectrum

MODULE – II: Adhoc Networks

2.1 Understand wireless ad-hoc networks concepts

- 2.1.1 Differentiate between Cellular and Adhoc Wireless networks
- 2.1.2 Discuss issues in Adhoc wireless networks
- 2.1.3 Explain the medium access schemes in wireless ad-hoc networks
- 2.1.4 Restate the QoS concepts and issues in wireless ad-hoc networks
- 2.1.5 Describe different MAC issues in wireless networks
- 2.1.6 Explain different requirements of routing protocol
- 2.1.7 Differentiate routing protocol issues and multicast routing protocol issues
- 2.1.8 List out different QoS parameters in wireless networks
- 2.1.9 What are different types of attacks in wireless networks
- 2.1.10 Discuss major security threats in wireless networks
- 2.1.11 List out different issues in energy management of wireless networks
- 2.1.12 State different deployment considerations in wireless networks
- 2.1.13 Describe performance degradation factors in transport layer for wireless network

MODULE – III: Adhoc Routing

3.1 Comprehend the MAC protocols in Ad-hoc networks

- 3.1.1 Explain the issues in designing MAC protocols
- 3.1.2 Classify the different MAC protocols.
- 3.1.3 Compare the various MAC protocols – MACAW, BTMA, MACA/PR-DPS
- 3.1.4 Discuss issues in designing MAC protocol for wireless networks

- 3.1.5 What are design goals for MAC protocol
- 3.1.6 List out different classifications of MAC protocol
- 3.1.7 How priority scheduling is implemented in MAC protocols

3.2 Understand the routing process and protocols in Ad-hoc networks

- 3.2.1 Differentiate between the different types of routing protocols – proactive, reactive and hybrid
- 3.2.2 Describe the various routing protocols DSDV, AODV, CGSR and FSR
- 3.2.3 Compare the routing protocols in Ad-Hoc networks.
- 3.2.4 What are different issues in routing protocol
- 3.2.5 Compare DSDV and AODV
- 3.2.6 Explain different characteristics of cluster based routing protocol
- 3.2.7 How fish eye concept can be implemented in a routing protocol

MODULE – IV: Transport layer

4.1 Understand the transport layer issues in ad-hoc networks

- 4.1.1 Explain the Issues in transport layer
- 4.1.2 Describe ATCP
- 4.1.3 Explain the functions of transport layer
- 4.1.4 Describe design goals of transport layer
- 4.1.5 Compare ATCP and FTCP
- 4.1.6 Why traditional TCP protocol cannot be used in wireless networks
- 4.1.7 Differentiate route failure notification and route reestablishment notification
- 4.1.8 What are four different states of ATCP

4.2 Understand wireless sensor networks

- 4.1.1 Define sensor networks
- 4.1.2 Describe the applications of Sensor networks
- 4.1.3 Compare sensor networks with Ad-hoc wireless networks
- 4.1.4 What are different subsystems in sensor networks
- 4.1.5 Explain different features of sensor nodes
- 4.1.6 What are advantages of sensor nodes
- 4.1.7 Describe Important operations of sensor nodes
- 4.1.8 Explain the architecture of mesh networks

CONTENT DETAILS

MODULE – I: Wireless Networks

Introduction to wireless LANs - IEEE 802.11 WLANs - Physical Layer- MAC sub layer- MAC Management Sub layer- Security- Wired Equivalent Privacy (WEP) and Wi-Fi Protected Access (WPA).

MODULE – II: Adhoc Networks

Introduction – Difference between Cellular and Adhoc wireless networks—Applications of Adhoc Wireless networks--Issues in Ad-Hoc Wireless Networks-Medium access scheme – Routing- Multicasting- TPC protocol-Pricing scheme- QoS- Self-organization-Security-Energy management-Addressing and service discovery-Deployment considerations

MODULE – III: Adhoc Routing

Issues in designing MAC protocol- Design goals of a MAC protocol-Classification of MAC protocol—MACAW-BTMA-MACA/PR-DPS-Issues in routing protocols – Characteristic of Ideal routing protocol-Classifications of routing protocols-proactive-reactive-hybrid-routing protocols-DSDV-AODV-CGSR-FSR

MODULE – IV: Transport layer

Adhoc TCP and other networks-Issues in transport layer protocols-Design goals-Feedback based TCP-ATCP-Sensor Network –Introduction-Applications-Comparison with Adhoc Wireless Networks-Mesh Networks

TEXT BOOK(S)

1. C.Siva Ram Murthy and B.Smanoj, — Ad Hoc Wireless Networks – Architectures and Protocols, Pearson Education, 2004.

REFERENCE

1. Feng Zhao and Leonidas Guibas, —Wireless Sensor Networks, Morgan Kaufman Publishers, 2004.
2. C.K.Toh, —Ad Hoc Mobile Wireless Networks, Pearson Education, 2002.